

PLATFORM & DISTRIBUTED SYSTEMS

Distributed Systems

Real-time Streaming

Apache Beam / Dataflow

Pub/SubKubernetes / GKE

IstioGitOps · ArgoCD

Infrastructure as Code

System Design

Low-latency Services

AI PLATFORM & LLM INFRASTRUCTURE

LLMsLoRA / QLoRAPEFT

UnslothvLLMTransformers

Fine-tuningModel Evaluation

LLM Deployment

ML & DEEP LEARNING

PyTorchTensorFlow

XGBoostScikit-learn

RAPIDS (cuDF/CuPy)

NVIDIA GPU Platforms

Computer VisionAutoencoders

Distributed Training

Feature Engineering

INFRASTRUCTURE & DELIVERY

Cloud & Orchestration

GCP Vertex AIGKEDataflow

GCSAzure ML Studio

Serving & Packaging

KubernetesHelmDocker

Seldon CoreMLflow

Model Registry

Observability & CI/CD

GrafanaPrometheus

Cloud BuildArgoCD

Google KCC

LANGUAGES & ARCHITECTURE

PythonSQL

Software Architecture

Testing & Debugging

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization.

WORK EXPERIENCE

- **Lead Artificial Intelligence Engineer**

Best Buy · Bengaluru, India

Jul 2025 - Present

 - Built batch, streaming and real-time fraud systems: Beam/Dataflow feature pipelines at **40,000 events/sec at peak**, secure, monitored GKE/Seldon Core inference and load testing. ML feeds a rules-based engine; combined fraud prevention saves **US\$1.5M/year**.
 - Established GitOps as reusable internal platform tooling — Cloud Build for CI, ArgoCD for CD, Google KCC for infrastructure-as-code — standardising how teams ship to GKE with autoscaling, Istio routing, and blue-green releases.
 - Built reusable LLM fine-tuning, evaluation, and shared vLLM/GKE serving with estimated annual API-cost savings of **~\$300K**. Kept inputs company-managed and reduced external API exposure during peak demand.
 - Lead architecture and production delivery across search, personalization, fraud, and cybersecurity; partner with data scientists to turn validated models into secure, monitored services.
- **Lead ML Engineer - ML Engineering & Operations**

Wipro · Client: Major US Retail Corporation

Aug 2023 - Jul 2025

 - Led ML/MLOps engineers and Morpheus/RAPIDS model lifecycles on **NVIDIA DGX (8xA100)**, training per-entity autoencoders on **billion-row CloudTrail datasets** with continuous evaluation, debugging, and Grafana alerting. Completed **300+ résumé screens** and **50+ technical interviews** at Wipro.
 - Launched phishing classification at **2,400 emails/sec - 2x prior throughput** and OCR-based abuse-inbox triage that doubled case volume, integrating predictions with Microsoft Sentinel and security-response workflows.
- **Senior Data Scientist — Data Science / Azure ML**

Accenture

May 2021 - Sep 2023

 - Led cross-functional architecture and launch of an Azure ML quality-prediction product: **\$2.5M annual savings**, inspections **130 → 160/day**, **+17%** defect detection, and **+20%** customer approval.
 - Operationalized data preparation, optimization, retraining, monitoring, and CI/CD; designed OR-Tools services, mentored engineers, and coordinated business, data science, platform, and client stakeholders.

EARLIER EXPERIENCE

Data Scientist
GD Research Centre Pvt Ltd
May 2020 - May 2021

Designed and deployed Python forecasting pipelines across 15+ industry verticals, from heterogeneous data preparation through client-facing results.

Deep Learning Engineer
Tata Consultancy Services
Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours; 90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML
Infosys
May 2016 - Jul 2019

Delivered design, coding, unit/integration testing, and production support; transitioned to Python ML in 2018 and shipped lead-conversion prediction.

CERTIFICATIONS

Oracle Cloud Infrastructure 2024
Generative AI Certified Professional

Google Cloud
Certified Generative AI Leader

Microsoft Certified
Azure AI Engineer Associate

Microsoft Certified
Azure Data Scientist Associate

EDUCATION

B.E. Electrical & Electronics
Andhra University, Visakhapatnam
2012 - 2016 · 80%

OPTIMISATION & SIMULATION

Google OR-Tools

Vehicle Routing

Mixed-Integer Programming

Constraint Modelling

Operations Research

Supply-Chain Optimization

MODEL LIFECYCLE

Experiment Tracking

Automated Retraining

Vertex AI Pipelines

Azure ML Pipelines

Vertex AI Feature Store

PRODUCTION ENGINEERING

ML System Architecture

Cloud-Native ML Systems

JWT Authentication

Load Testing

Unit / Integration Testing

ENGINEERING LEADERSHIP

Architecture Reviews

Technical Mentoring

Technical Hiring Team Building

Stakeholder Alignment

Cross-Functional Delivery

PROJECTS

Real-time Fraud & Abuse Detection Platform

Best Buy · Aug 2025 - Jan 2026

Large-scale Beam/Dataflow pipelines transform clickstream and transaction events into Vertex AI Feature Store signals, scoring abuse across payments, refunds, reships, chargebacks and "item not received" fraud. XGBoost models deploy through Seldon Core on GKE with JWT security, load testing, GitOps releases, and production monitoring.

Apache Beam Vertex AI

40k feature events/sec · US\$1.5M/year combined fraud savings · Full lifecycle deployment

AWS Account Behavioral Fingerprinting

Wipro · Jan 2024 - Aug 2025

Per-entity autoencoders learn normal AWS account, user, service, and machine activity. Morpheus/RAPIDS training and inference run on DGX (8xA100) over billion-row CloudTrail datasets with continuous retraining and Grafana alerts.

RAPIDS / cuDF

Billion-row datasets · Continuous model lifecycle · 8xA100 DGX

Proofpoint Phishing Classification

Wipro · Oct 2023 - Feb 2024

Built Morpheus training and inference pipelines on DGX for preprocessing, feature engineering, model evaluation, and live scoring. Integrated security-team feedback and Microsoft Sentinel data to catch phishing that bypassed initial filters.

RAPIDS DGX Microsoft Sentinel

2,400 emails/sec · 2x prior throughput · Production security workflow

Abuse-Inbox Triage & Brand-Infringement Takedown

Wipro · 2024 - 2025

GPU-accelerated OCR pipeline extracting scammer phone numbers and email addresses from multiformat abuse reports — text, HTML, image email, and PDF/PNG/JPEG attachments, including colour-obfuscated and whitespace-split digits. Identifiers auto-route to Google and iCloud for brand-infringement takedowns and to the security response team.

pytesseract OCR Internal Tooling Abuse Response

2x prior case volume · Automated routing to takedown & security teams

LLM Fine-tuning, Eval & Serving Platform

Best Buy · Jul 2025 - Present

Config-driven Python platform for LoRA/QLoRA fine-tuning, data preparation, multi-task training, automated evaluation, artifact management, and vLLM serving. Supports Unsloth and PEFT/Transformers with Accelerate; adapters hot-load into a shared GKE deployment.

~\$300K estimated annual API-cost savings

OPEN SOURCE & PLATFORM BUILDING

GitOps Delivery Platform

Best Buy · Cloud Build · ArgoCD · Google KCC

Internal developer platform for multiple product teams — every model, pipeline and GCP resource declared in Git, with zero-downtime ArgoCD releases to GKE.